

El Niño and the California Water Year: A Descriptive Analysis of ENSO Intensity and Statewide Precipitation

DATASCI 203 Final Project (Group 2)

Daniel Bowd, Gunhui Kim, Rishi Raj, and Sasha Schaps

August 4th, 2026

Introduction

Rainfall in California varies from year to year with weather patterns in the tropical Pacific. The El Niño–Southern Oscillation (ENSO), a coupled pattern of tropical Pacific ocean temperatures and air pressure, is the one climate signal widely believed to foreshadow the state’s wet and dry years.

This report asks the following research question.

What is the relationship between the intensity of ENSO and total statewide precipitation in California over the corresponding water year?

Describing this relationship is valuable on its own terms. It tells the public and planners in government and utilities (the Bureau of Reclamation, the California Department of Water Resources, and PG&E) what history actually says about ENSO. Prior research documents the qualitative pattern, in which El Niño tends to bring wetter-than-normal conditions to California and La Niña the inverse (Cayan, Redmond & Dettinger 1999, Redmond & Koch 1991, Schonher & Nicholson 1989), with stronger associations in Southern California (Jong, Ting & Seager 2016). Yet the relationship is famously non-deterministic, and recent years were unexpectedly wet despite La Niña (Guirguis et al. 2024). Our contribution is to put statewide, water-year-level numbers on that pattern. We deliberately make no causal or predictive claims.

Data

Provenance

- **California statewide precipitation** (`precip_in`): Total California water-year (October–September) precipitation, in inches. Our intended source was NOAA’s Climate at a Glance statewide series, but that service was returning HTTP 503 throughout the analysis period. We therefore construct the outcome directly from NOAA GHCN-Daily station records: for each water year we total each station’s monthly precipitation, keep only stations reporting all twelve months, and average across the 198 stations (median) available that year. This is a station average rather than an area-weighted statewide value, so its level runs roughly one inch higher than the published statewide figure; the ENSO relationship this report describes is materially unchanged (correlation with winter ONI of 0.243 against 0.257 for the published series). The pipeline prefers the official file automatically if it is placed in `data/external/`.
- **Oceanic Niño Index** (`oni_djf`): NOAA’s December–February Oceanic Niño Index (ONI), averaged within each water year. Phase indicators, strength categories, a one-year lag, and a time trend were derived from this series.

- **Southern Oscillation Index** (`rsoi`): November–March SOI multiplied by -1 so that positive values correspond to El Niño conditions.

Index calculations. ENSO intensity is measured using NOAA’s Oceanic Niño Index (ONI), defined as the December–February average sea-surface temperature anomaly in the Niño 3.4 region. We also retain a reversed Southern Oscillation Index (SOI) as an alternative operationalization. Detailed definitions are provided in the appendix.

Operationalization

Precipitation concept → `precip_in`. Statewide water-year totals are the standard measure for “how wet was California’s year,” and the water year cleanly contains one winter storm season.

ENSO intensity concept → `oni_djf`. We make use of the Oceanic Niño Index (ONI) as the ENSO measure because it is NOAA’s operational ENSO indicator and represents intensity as a continuous sea surface temperature anomaly (°C). ENSO categories are used not as model predictors, but for visualizations. All 76 water years have complete observations, Model 3 using the previous winter ONI and therefore has `n` equaling 75 due to the missing value for WY1950.

The Shape of the Data

The rest of the report describes the joint distribution of winter ENSO intensity and water-year precipitation, starting from the raw ingredients. Table 1 shows the statistics for the outcome and the ENSO measures.

Table 1: Descriptive statistics for the analysis variables, WY 1950–2025. The lagged ONI has 75 observations because it is undefined for the first sample year.

Variable	N	Mean	SD	Min	Median	Max
Precipitation (in)	76	23.13	7.71	11.47	21.17	44.77
ONI (Dec-Feb, °C)	76	0.01	1.04	-1.80	-0.15	2.60
Previous winter ONI (°C)	75	0.02	1.04	-1.80	-0.10	2.60

The full 76-year history complicates the expectation that El Niño years are wet years, because the wettest columns below are not all El Niño. The record winters of 1983 and 1998 were very strong El Niños, but 2017, the fourth-wettest year, was Neutral. The very strong El Niño winter of 2016 coincided with a roughly average year, and the second-driest year on record, 1977, was itself a weak El Niño.

Table 2: Statewide water-year precipitation by ENSO phase. El Niño years average 3 or more inches wetter but are also markedly more variable.

ENSO phase (ONI, DJF)	Water years	Mean precip (in)	SD (in)	Driest (in)	Wettest (in)
La Niña	29	22.3	6.2	11.5	36.4
Neutral	20	21.5	7.4	12.8	37.6
El Niño	27	25.2	9.1	11.5	44.8

A simple comparison of group means offers only partial support for the expectation that El Niño years are wet years. On average, El Niño years received 3.3 more inches of precipitation than the other years

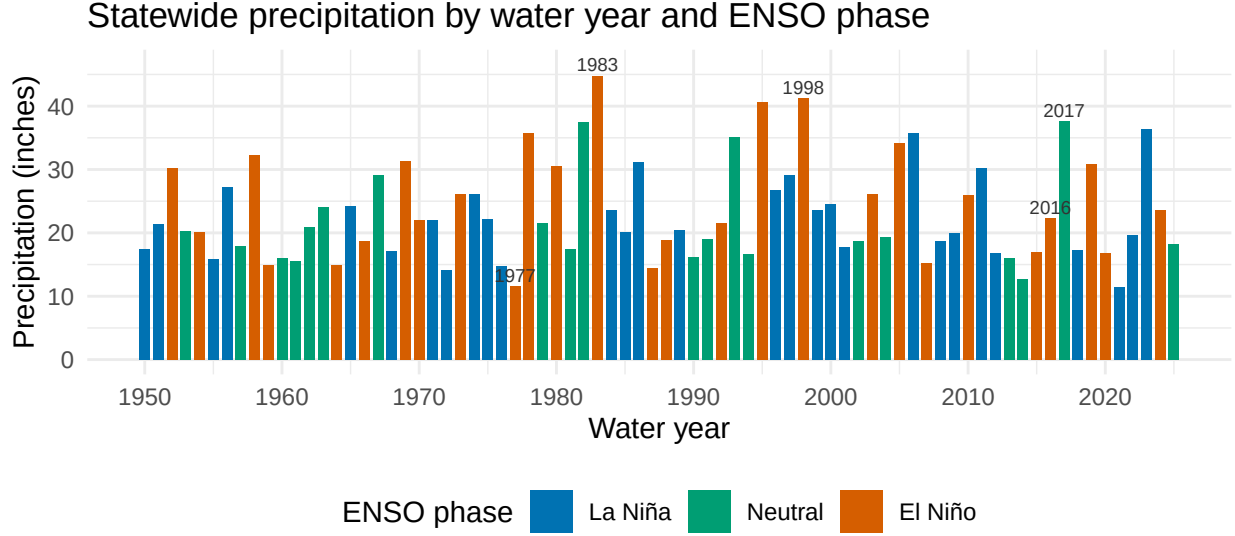


Figure 1: California statewide water-year precipitation, 1950–2025, colored by ENSO phase. The wettest years are not uniformly El Niño years, previewing the noisy relationship the models quantify.

taken as a group, but the difference is not statistically significant at the 5% level (Welch two-sample t-test, $|t| = 1.66$, two-sided $p = 0.105$). Because collapsing ENSO into categories discards information about event intensity, the regression models below work with the full metric index instead. Table 2 also shows that El Niño years are the most *variable* (SD 9.1 inches, versus roughly 6.8 for the other phases), a first sign of the heteroskedasticity the models will need to address.

The ENSO–Precipitation Relationship

Regression Models

We estimate four models. Each one asks a different question of the same 76 water years, and reading them together is what separates a robust description from an artifact of one specification.

Model	Specification	Question it answers
1	<code>precip_in ~ oni_djf</code>	Is there a linear ENSO–precipitation association?
2	<code>precip_in ~ el_nino + la_nina</code>	What does collapsing intensity into categories cost?
3	<code>precip_in ~ oni_djf + oni_djf_lag1 + time_trend</code>	Is the association confounded by last winter’s ENSO or by trend?
4	<code>log(precip_in) ~ oni_djf</code>	Is the relationship proportional rather than additive?

Model 1 is the headline. A one-degree increase in winter ONI is associated with 1.81 additional inches of water-year precipitation (robust SE 0.88, $p = 0.044$), but it accounts for only 5.9% of the variation in how wet California’s year is.

The remaining models each reinforce that reading. Model 2 shows what categories cost: neither the El Niño indicator (3.74 inches, $p = 0.124$) nor the La Niña indicator reaches significance, and R^2 falls to 0.044 – discarding intensity discards real signal. Model 3 shows the association is not an artifact of persistence or trend: the contemporaneous ONI coefficient is essentially unchanged at 1.88 ($p = 0.046$) once the previous winter’s ONI and a linear time trend are included, and the trend itself is flat ($p = 0.636$). Model 4 restates the relationship proportionally: each degree of ONI corresponds to roughly 6.7% more precipitation ($p = 0.060$).

Regression Assumptions

1. IID sampling. Water years are not drawn independently: ENSO is a physical system with memory,

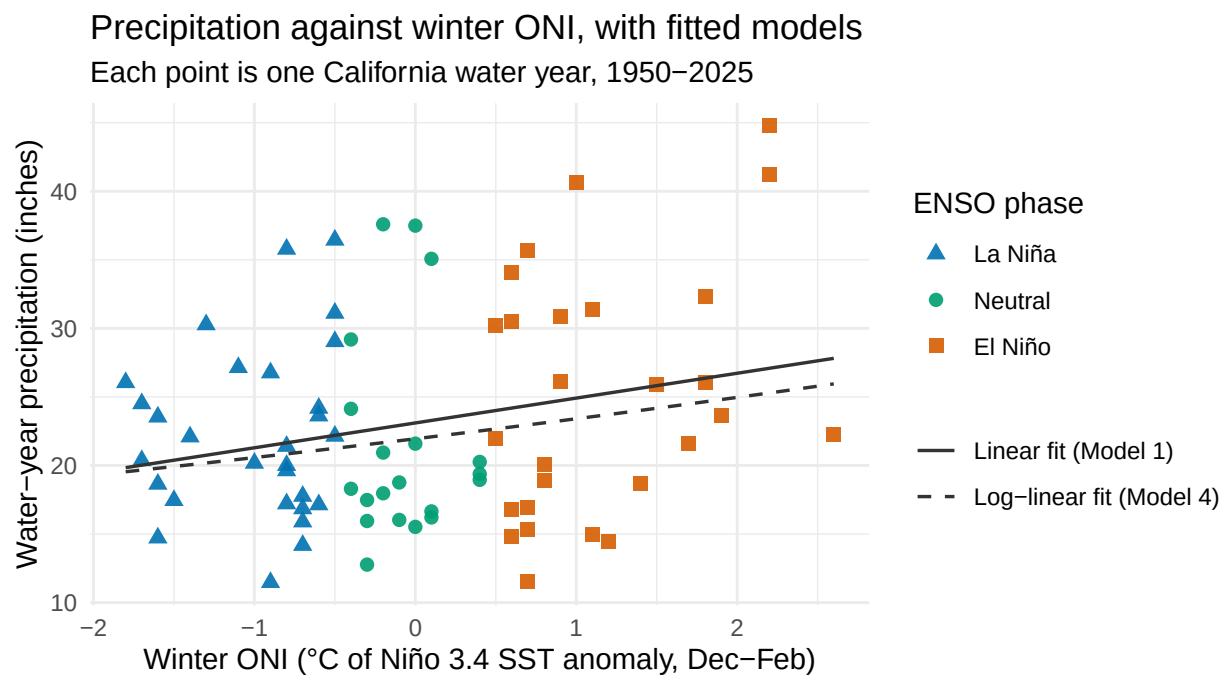


Figure 2: California water-year precipitation against winter ONI, WY 1950–2025, colored and shaped by NOAA ENSO phase. The solid line is the simple OLS fit (Model 1) and the dashed curve is the log-linear fit (Model 4) transformed back to inches (a geometric-mean prediction, slightly below the arithmetic mean). Both show a modest upward relationship, and the fanning of points at high ONI shows El Niño years are more variable.

Table 3: ENSO intensity and California water-year precipitation. Models 1-3 are in inches; Model 4's outcome is log inches, so its coefficient is a proportional effect and its R-squared is not comparable to the others.

	<i>Dependent variable:</i>			
	Precipitation (in)		log(Precipitation)	
	M1	M2	M3	M4
Winter ONI	1.812** (0.883)		1.878** (0.926)	0.064* (0.034)
El Nino		3.738 (2.405)		
La Nina		0.749 (2.016)		
Previous winter ONI			1.302* (0.774)	
Time trend			0.018 (0.039)	
Constant	23.101*** (0.861)	21.512*** (1.652)	22.418*** (1.546)	3.089*** (0.036)
Observations	76	76	75	76
R ²	0.059	0.044	0.088	0.044
Adjusted R ²	0.046	0.018	0.049	0.031

Note:

*p<0.1; **p<0.05; ***p<0.01

HC1 heteroskedasticity-robust standard errors.

4. Homoskedastic errors. This assumption **fails** in levels. El Nino years are markedly more variable than other years (Table 2), and a Breusch–Pagan test rejects homoskedasticity ($p = 0.015$). We therefore report **HC1 robust standard errors throughout**, which is why the Model 1 standard error is 5% larger than the classical one. The log transform largely resolves the problem ($p = 0.135$), which is an independent reason to report Model 4.

5. Normally distributed errors. Precipitation is right-skewed (skewness 0.83), and Shapiro–Wilk rejects normality of the Model 1 residuals ($p = 0.001$). The log transform pulls the outcome close to symmetric (skewness 0.26; Shapiro–Wilk $p = 0.081$). With $n = 76$ the central limit theorem gives some protection, but this is a genuine limitation and the reason we treat Model 4’s proportional statement as the more trustworthy of the two functional forms.

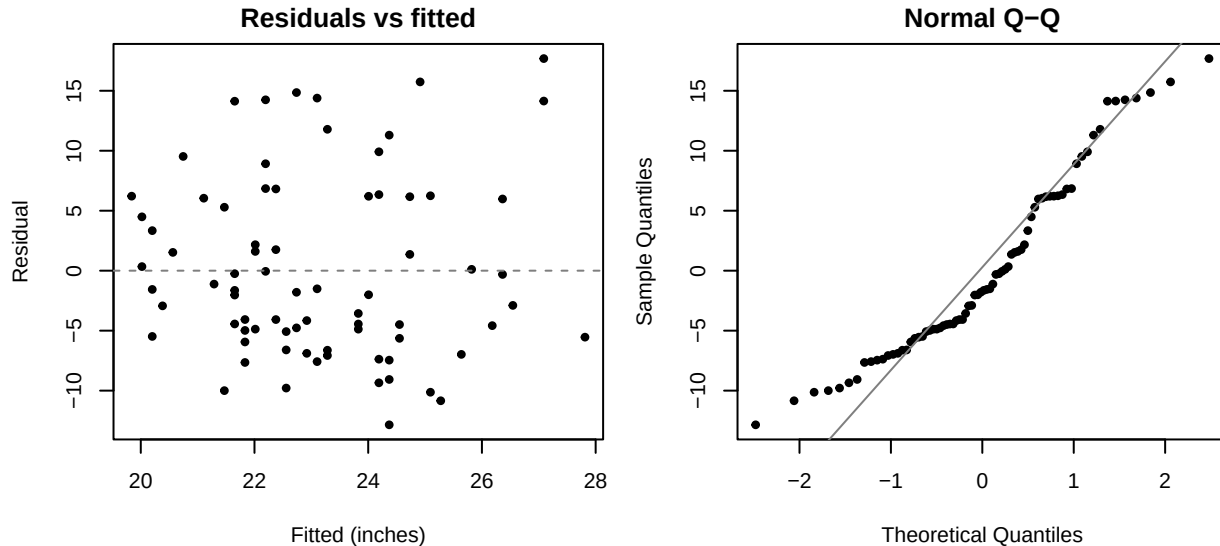


Figure 3: Model 1 diagnostics. Left: residuals against fitted values, showing the wider spread at high fitted values that the Breusch–Pagan test detects. Right: normal Q–Q plot, with departures in both tails reflecting the right-skewed outcome.

Results and Practical Significance

Statistical significance. The association between winter ENSO intensity and California water-year precipitation is positive and statistically significant at the 5% level, but only just: Model 1 gives $p = 0.044$ with robust standard errors, and the proportional version in Model 4 sits just outside conventional significance ($p = 0.060$). The finding survives the controls in Model 3 but does not survive being collapsed into categories in Model 2. A reader should conclude that the relationship is real and consistently signed, not that it is firmly established.

Table 4: Effect sizes implied by Model 1. The last column is the one that matters for planning: it compares the ENSO signal against the year-to-year noise it must be detected within.

Scenario	Change (in)	As % of mean	As % of SD
One SD increase in winter ONI	1.9	8.1	24.3
Strong El Nino (ONI = +1.5)	2.7	11.8	35.2
Strong La Nina (ONI = -1.5)	-2.7	-11.8	-35.2

Practical significance. The effect is real but small relative to the noise a water manager actually faces. A one-standard-deviation swing in winter ONI moves expected precipitation by 1.9 inches – 8% of the average year, but only 24% of a standard deviation of annual precipitation. Put the two extremes side by side: the expected gap between a strong El Nino and a strong La Nina is about 5.4 inches, against a year-to-year standard deviation of 7.7 inches. ENSO shifts the distribution; it does not determine the outcome.

That asymmetry is the practical message. Table 2 makes it concrete: El Nino years average 25.2 inches against 22.3 for La Nina years, yet the wettest La Nina year on record still exceeded the average El Nino year. Planning that treats an El Nino forecast as a guarantee of a wet year is not supported by 76 years of evidence.

Conclusion

Across four specifications, winter ENSO intensity is positively associated with California water-year precipitation. The central estimate is 1.81 inches per degree of winter ONI (equivalently about 7% more precipitation per degree), it is statistically significant at the 5% level under robust standard errors, and it is unchanged by controls for the previous winter’s ENSO state and for trend.

The more useful finding, though, is how *little* of California’s hydrological variability this explains – 5.9% in Model 1. The historical record contains very strong El Nino winters that produced ordinary years and neutral winters that produced near-record ones. For the agencies that motivated this question, the honest summary is that ENSO is a genuine but weak tilt in the odds: useful as one input among many for water-supply planning, and not sufficient on its own to anticipate a wet or dry year.

We make no causal claim. ENSO is not randomly assigned, our sample is a single realization of one climate system observed over 76 years, and the outcome is a statewide aggregate that averages over the strong regional differences documented in prior work (Jong, Ting & Seager 2016). What this report establishes is the shape and size of a historical association, which is what a descriptive analysis can honestly deliver.

Appendix

A.1 References

- Cayan, D. R., Redmond, K. T., and Dettinger, M. D. (1999). ENSO and hydrologic extremes in the western United States. *Journal of Climate*.
- Guirguis, K., et al. (2024). Reinterpreting ENSO’s role in modulating impactful precipitation events in California. *Geophysical Research Letters*.
- Jong, B.-T., Ting, M., and Seager, R. (2016). El Niño’s impact on California precipitation: seasonality, regionality, and El Niño intensity. *Environmental Research Letters*.
- NOAA National Centers for Environmental Information. *El Niño/Southern Oscillation (ENSO): Southern Oscillation Index (SOI) calculation*. <https://www.ncei.noaa.gov/access/monitoring/enso/soi> (accessed August 2026).
- Redmond, K. T., and Koch, R. W. (1991). Surface climate and streamflow variability in the western United States and their relationship to large-scale circulation indices. *Water Resources Research*.
- Schonher, T., and Nicholson, S. E. (1989). The relationship of California rainfall to ENSO events. *Journal of Climate*.

A.2 Data Sources

The whole pipeline is reproducible from a single command, `Rscript src/data/make_analysis_data.R`, which rebuilds every input from source and writes the one file this report reads.

- **Precipitation.** The intended source was NOAA NCEI, *Climate at a Glance, Statewide Time Series*, as 12-month totals ending September (<https://www.ncei.noaa.gov/access/monitoring/climate-at-a-glance/statewide/time-series>). That service returned HTTP 503 throughout the analysis period, so `src/data/make_ca_wateryear_precip.R` falls back to a composite built from NOAA GHCN-Daily station records (`data/external/ca_monthly_rainfall.csv`), summing each station’s twelve monthly totals per water year and averaging across stations. The script prefers the official series automatically whenever `data/external/data_rainfall.csv` is present, and the `precip_source` column in the analysis file records which was used.
- **Oceanic Niño Index (ONI).** Read directly from the NOAA Climate Prediction Center table at <https://www.cpc.ncep.noaa.gov/data/indices/oni.ascii.txt>, preserved raw at `data/external/oni_download.txt` and reduced to one row per water year by `src/data/make_oni_djf_wateryear.R` (`data/interim/oni_djf_wateryear.csv`). We read CPC directly rather than via the `rsoi` package because `rsoi` recomputes ONI as a rolling three-month mean and therefore cannot produce the DJF window for WY1950.
- **Southern Oscillation Index (SOI).** From the NOAA Climate Prediction Center, <https://www.cpc.ncep.noaa.gov/data/indices/soi>, preserved raw at `data/external/soi_download.txt` and reduced to the Nov–Mar reversed-SOI series by `src/data/make_soi_wateryear.R` (`data/interim/soi_ndjfm_wateryear.csv`).
- **GHCN-Daily station metadata**, the measurement network behind the precipitation composite, comes from NOAA NCEI, <https://www.ncei.noaa.gov/pub/data/ghcn/daily/>, with the California station list at `data/external/ca_station_list.csv`.
- **Merge and validation.** `src/data/make_analysis_data.R` joins the three interim files on water year and refuses to write unless the result is a complete consecutive run of 76 water years with no missing values in either analysis variable, mutually exclusive phase indicators, and exactly one missing lagged ONI (WY1950). It writes the canonical `data/processed/caenso_analysis.csv`.
- **Data dictionary.** `references/data_dictionary.csv` (rendered as `references/datadictionary.png`) documents every variable’s type, units, range, source, and derivation, including all thirteen columns of the canonical analysis file.

A.5 Data Dictionary

Table 5: Data dictionary for the canonical analysis file (from Data/data_dictionary.csv).

Variable	Type	Units	Range / levels	Description
water_year	integer	water year	1950 to 2025	California water year (Oct 1 of prior calendar year through Sep 30). The unit of observation; one row per water year.
precip_in	numeric	inches	11.47 to 44.77	Outcome. Total statewide water-year precipitation, inches. Built by src/data/make_ca_wateryear_precip.R.
oni_djf	numeric	deg C	-1.8 to 2.6	Primary X. Oceanic Nino Index averaged over Dec-Feb of the water year, from NOAA CPC. Positive = El Nino.
oni_djf_lag1	numeric	deg C	-1.8 to 2.6	Previous water year oni_djf. Used in Model 3; NA for WY1950 (no prior year in sample).
rsoi	numeric	index (unitless)	-3.88 to 4.76	Alternative X. Nov-Mar Southern Oscillation Index multiplied by -1 so positive = El Nino. NA for WY1950-1951 (CPC SOI starts 1951).
enso_phase	character	category	La Nina; Neutral; El Nino	ENSO phase from oni_djf using NOAA cutoffs: El Nino if ≥ 0.5 , La Nina if ≤ -0.5 , else Neutral.
phase	character	category	La Nina; Neutral; El Nino (NA for WY1950-1951)	Second, independent phase classification: the same ± 0.5 rule applied to rsoi (atmosphere) rather than oni_djf (ocean).
enso_strength	character	category	neutral; weak; moderate; strong; very_strong	Intensity bin from oni_djf : neutral < 0.5 , weak < 1.0 , moderate < 1.5 , strong < 2.0 , very_strong ≥ 2.0 .
el_nino	integer	0/1 indicator	0; 1	Indicator equal to 1 when enso_phase is El Nino. Model 2 predictor.
la_nina	integer	0/1 indicator	0; 1	Indicator equal to 1 when enso_phase is La Nina. Model 2 predictor.
time_trend	integer	years since WY1950	0 to 75	Linear time trend, 0 at WY1950. Model 3 control for secular drift.
n_stations	integer	stations	153 to 226	Number of GHCN stations contributing a complete 12-month record that water year (fallback source only).
precip_source	character		single value; see description	Provenance flag naming the precipitation source actually used, so a reader can tell the official statewide series from the station composite.
id	character		284 unique values (e.g., USC00040029; USC00040136; USC00040144)	GHCN-Daily station identifier (e.g., USC00040029); joins to ca_station_list.csv.

Variable	Type	Units	Range / levels	Description
ym	character	date (YYYY-MM-01)	1905 unique values (e.g., 1867-06-01; 1867-07-01; 1867-08-01)	Calendar month of observation, stored as the first day of the month.
prcp	numeric	mm	0.00 to 1373.50	Total monthly precipitation summed from GHCN-Daily station records.
n_days	integer	days	25 to 31	Number of days in the month with a reported observation; months with fewer than 25 reporting days were dropped upstream.
region	character	category	NorCal; SoCal	California region the stations were aggregated into: NorCal or SoCal.
ym	character	date (YYYY-MM-01)	951 unique values (e.g., 1867-11-01; 1867-12-01; 1868-01-01)	Calendar month of observation, stored as the first day of the month.
prcp	numeric	mm	0.00 to 468.63	Mean monthly precipitation across reporting stations in the region.
n_stations	integer	count	1 to 168	Number of stations contributing to the regional mean that month; early years rely on very few stations.
oni	numeric	deg C	-2.03 to 2.75 (833 NA)	NOAA Oceanic Nino Index (3-month running SST anomaly, Nino 3.4 region); NA before the ONI record begins in 1950.
month	integer	1-12	1 to 12	Calendar month number, extracted from ym for seasonal modeling.
phase	character	category	ElNino; LaNina; Neutral	ENSO phase classification derived from ONI: ElNino, LaNina, or Neutral (Neutral where ONI is unavailable).
id	character		284 unique values (e.g., USC00040029; USC00040136; USC00040144)	GHCN-Daily station identifier; primary key, joins to ca_monthly_rainfall.csv.
lat	numeric	decimal degrees	32.57 to 41.99	Station latitude (WGS84).
lon	numeric	decimal degrees	-124.24 to -114.17	Station longitude (WGS84); negative values are west of the prime meridian.
element	character		PRCP	GHCN element code; PRCP for all rows (precipitation stations only).
first_year	integer	year	1867 to 1980	First year of precipitation records at the station.
last_year	integer	year	2024 to 2026	Most recent year of precipitation records at the station.
state	character		CA	US state code; CA for all rows.
name	character		284 unique values (e.g., ADIN RS; ALPINE; ALTADENA)	Station name as listed in GHCN-Daily metadata.
elev	numeric	meters	-59.10 to 2878.80	Station elevation above sea level; negative values are below sea level.
year	integer	water year	1950 to 2025	Water year label (Oct 1 of prior year through Sep 30), WY 1950-2025.
annual_precip_in	numeric	inches	7.93 to 42.46	Total California statewide water-year precipitation from NOAA Climate at a Glance (12-month totals ending September). The report's outcome variable (precip_in).

Variable	Type	Units	Range / levels	Description
water__year	integer	water year	1950 to 2025	Water year label, WY 1950-2025; merge key to the annual precipitation file.
oni_djf	numeric	deg C	-1.80 to 2.60	Oceanic Nino Index: Dec-Feb average SST anomaly in the Nino 3.4 region (NOAA CPC via <code>rsoi::download_oni()</code>). Primary ENSO intensity measure (Models 1, 3, 4).
enso_phase	character	category	el_nino; la_nina; neutral	NOAA ENSO phase for the water year based on winter ONI: el_nino (ONI $\geq$ 0.5), la_nina (ONI $\leq$ -0.5), or neutral.
enso_strength	character	category	moderate; neutral; strong; very_strong; weak	ONI-based intensity category: neutral, weak, moderate, strong, or very_strong; used for visualization, not as a model predictor.
el_nino	integer	0/1 indicator	0 to 1	Dummy equal to 1 in El Nino water years (Model 2 predictor).
la_nina	integer	0/1 indicator	0 to 1	Dummy equal to 1 in La Nina water years (Model 2 predictor).
oni_djf_lag1	numeric	deg C	-1.80 to 2.60 (1 NA)	Previous winter's ONI (one-year lag); NA for WY1950, the first sample year (Model 3 predictor).
time_trend	integer	index (0-75)	0 to 75	Linear time index, 0 for WY1950 through 75 for WY2025 (Model 3 trend control).